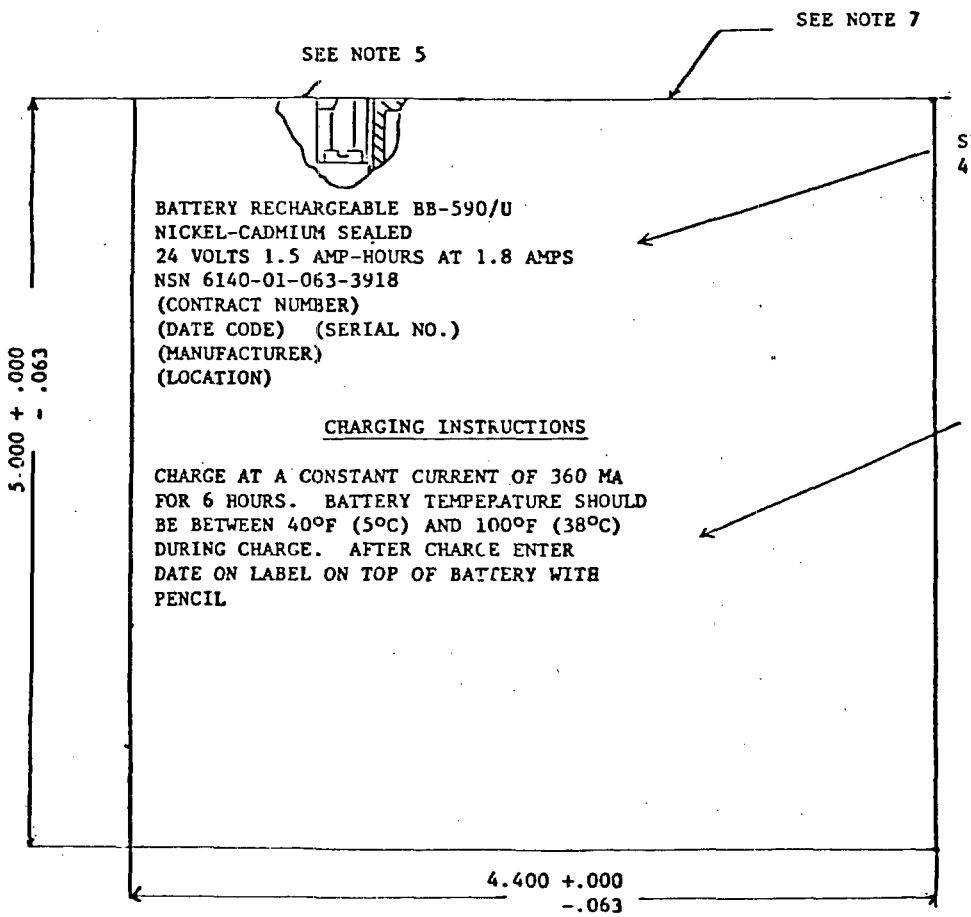
MILITARY SPECIFICATION SHEET

BATTERY, RECHARGEABLE, SEALED NICKEL CADMIUM BB-590/U

This specification sheet is approved for use by the Electronic Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the sealed nickel cadmium rechargeable battery type described herein shall consist of this document and the latest issue of Specification MIL-B-49436(ER).

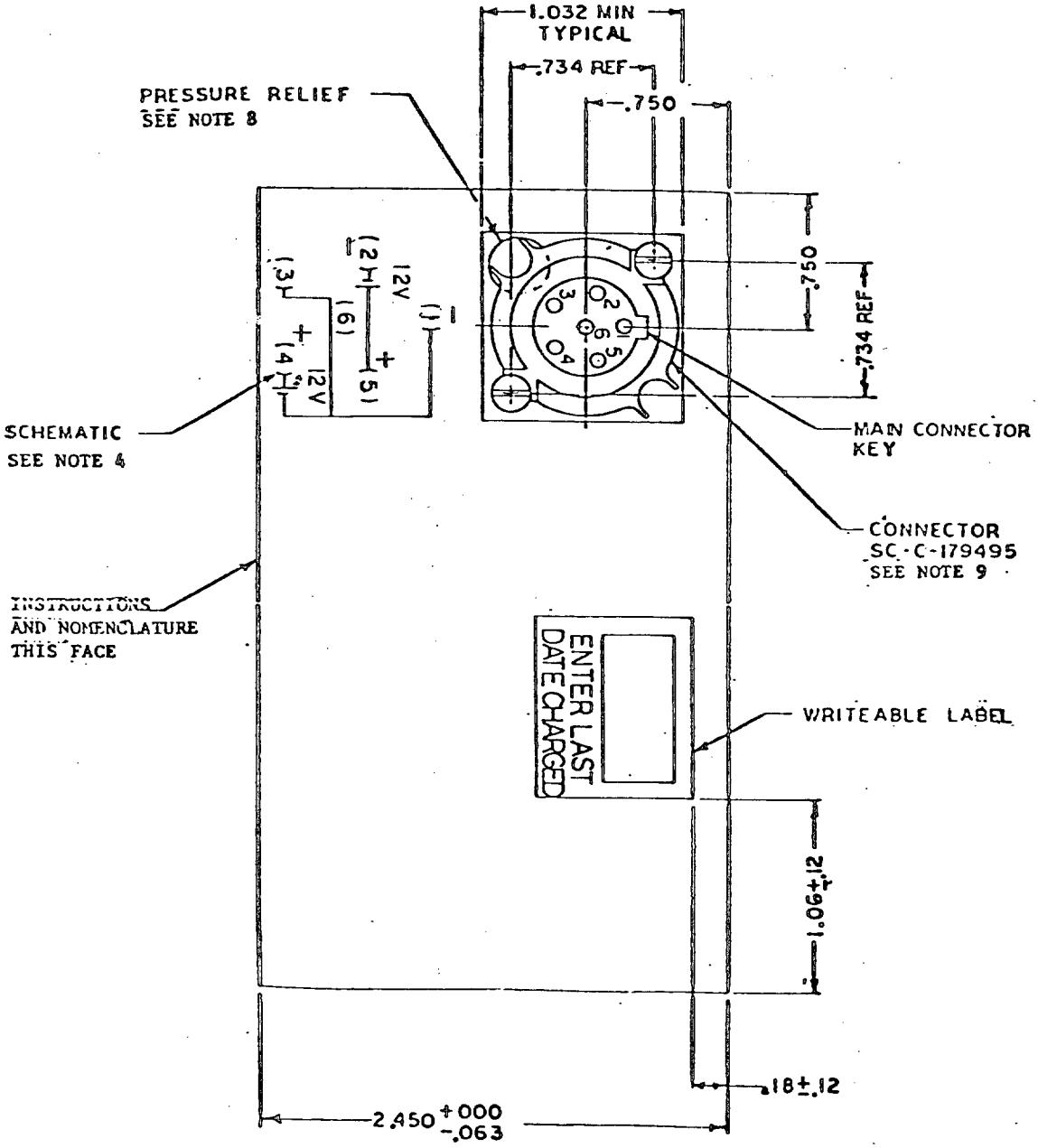


INCHES	MM
.063	1.60
4.400	111.76
5.000	127.00

FIGURE 1. FRONT VIEW

FSC-6140

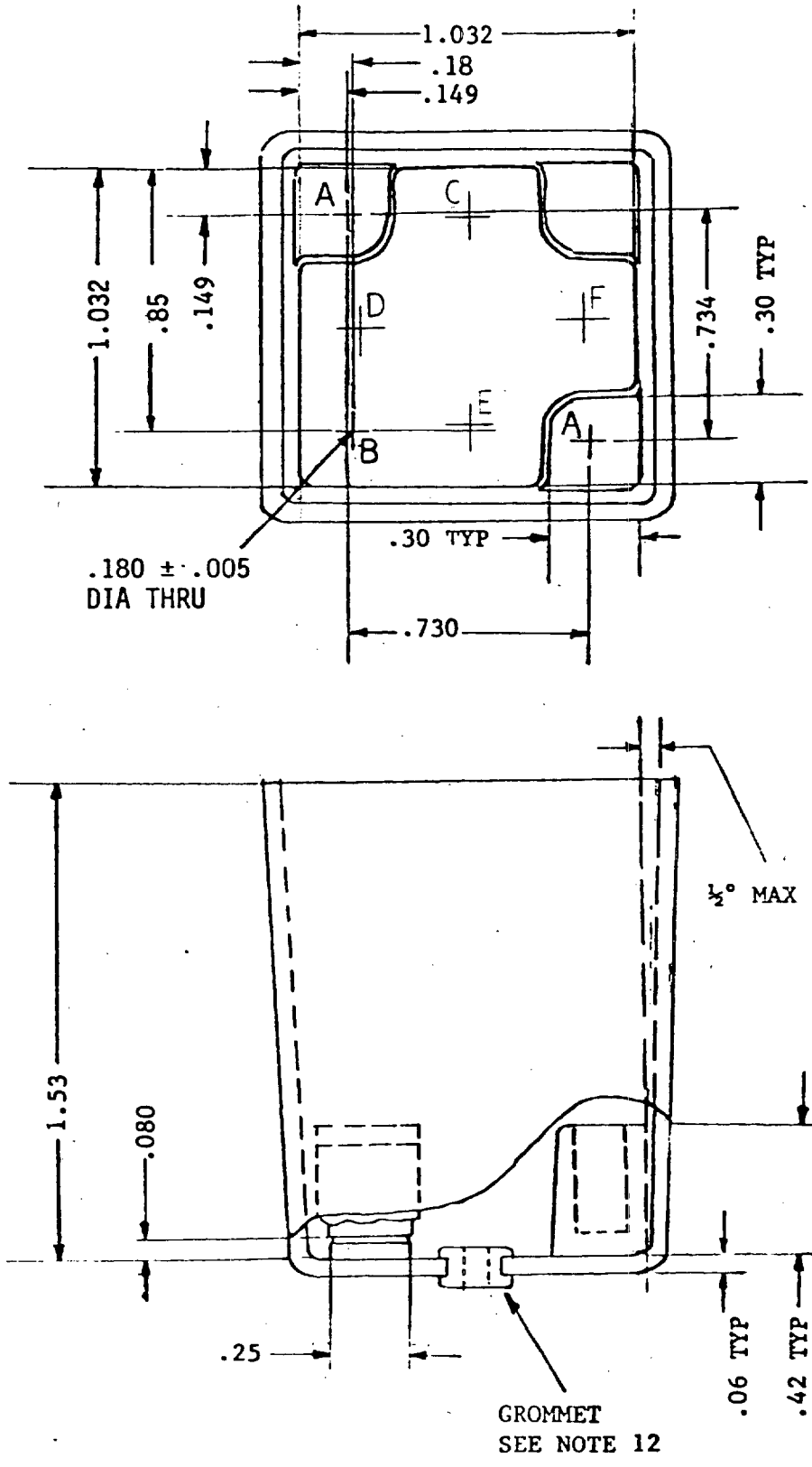
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INCHES	MM
.063	1.60
.120	3.05
.180	4.57
.734	18.64
.750	19.05
1.032	26.21
1.060	26.92
2.450	62.23

FIGURE 2. TOP VIEW

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INCHES	MM
.005	.13
.060	1.52
.080	2.03
.149	3.78
.180	4.57
.250	6.35
.300	7.62
.420	10.67
.730	18.54
.734	18.64
.850	21.59
1.032	26.21
1.530	38.86

FIGURE 3. BATTERY-CONNECTOR INTERFACE

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DRAWING NOTES

1. All dimensions in inches.
2. Tolerances on decimals ± .031 inches.
3. Dimensioning shall be per USAS-144.5 Y14.5 1966.
4. Stencil (or equivalent) schematic, identification and charging instructions as shown. Color marking white.
5. Face of connector shall be flush to .031 below top of battery.
6. Manufacturer must fill in applicable information in parenthesis.
7. See Figure 2 for top view.
8. Pressure relief valve shall be located under connector. It shall release gas before case damage is incurred.
9. See Figure 3 for connector-battery interface. The main connector key shall be oriented as shown in Figure 2. The two connector mounting screws shall be corrosion resistant steel, Class 303, Condition A in accordance with ASTM A 484.
10. Holes A shall be .154 ±.002 dia x .34 ±.02 deep for a 4-40 UNC-2A insert. Insert to be flush or below surface.
11. The lettered holes of table I are shown in approximate positions in Figure 3. The holes shall be .136 ±.002 diameter. Marking of holes not required.
12. Each required SC-C-179495 connector wire shall be provided with a hole (see table I and figure 3) and pre-fitted with a neoprene rubber grommet of 50 ±5 durometer in accordance with MIL-R-3065 to furnish a watertight seal around the connector wires.
13. Pressure relief valve mounted in hole B has a typical height of .42 from base on connector well.

TABLE I. CONNECTOR WIRE COLOR CODE

HOLE C	HOLE D	HOLE E	HOLE F
BLACK/WHITE	BLACK	WHITE/RED	RED

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REQUIREMENTS:

Rated capacity (C_5):	1.8 AH
Voltage (Nominal):	24.0 volts
Weight (Maximum):	1.82 Kg (4.0 lbs)
Connector:	Floating type per SC-C-179495
Number and type of cells:	20 each sealed "C_SD" size cells or equal in series
Charge rate for testing and on battery (C/5 for 6 hrs):	0.36 Amps

Discharge Rates	5 Second Voltage(V)	Final Voltage(V)	Current (Amps)
C/5	-	20.0	0.36
C_5	-	20.0	1.8
$5C_5$	22.0	20.0	9.0
$10C_5$	-	20.0	18.0

Test	Discharge Rate	Capacity Requirement(AH)
Full Capacity Discharge	C_5	1.5
Cycle Life, 224 cycles	C_5	1.2
Overcharge, 24 hours	C_5	1.5
High Rate Discharge	5 C_5	1.1
Low Temperature Discharge at -30°C (-22°F)	C_5	0.6
Retention of Charge 7 days at 51°C (124°F)	C_5	0.75
Pulse Discharge 18 amps, 5 seconds on, 25 seconds off, continuously to 20 volts	10 C_5	1.1
Vibration	C/5	1.8

NOTES - 1. All values are minimum, unless otherwise specified.
2. All charges and discharges shall be performed on fully assembled batteries through the terminals.

CUSTODIAN:
ARMY-ER
REVIEWER:
ARMY - CR, AV, GL

PREPARING ACTIVITY:
ARMY-ER
(PROJECT 6140-A577-1)

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